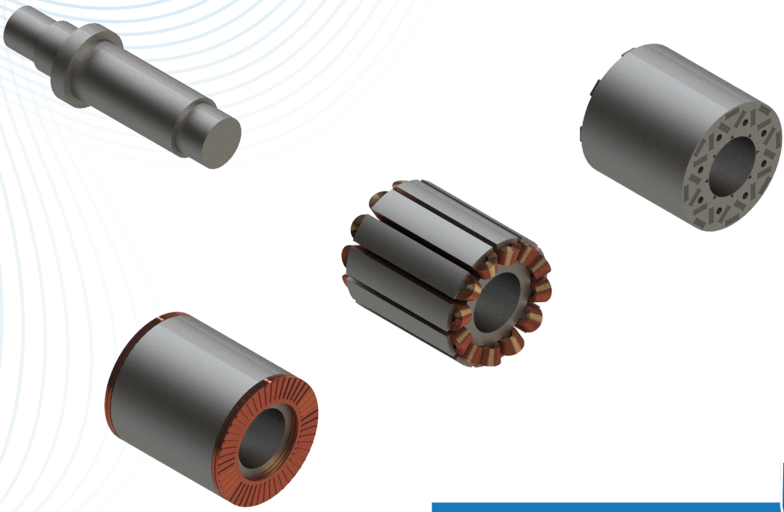

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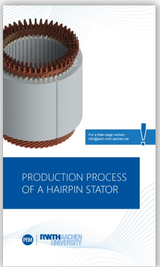
PRODUCTION PROCESSES OF ROTORS

PEM | RWTH AACHEN UNIVERSITY

Guides

to electric motor production

In its “Guide to Electric Motor Production” series, the Chair of Production Engineering of E-Mobility Components” (PEM) of RWTH Aachen University presents the unique process chains for manufacturing hairpin stators, continuous hairpin stators, rotors, and axial flux motors.



Production Process of a Hairpin Stator

The “Production Process of a Hairpin Stator” guide covers the entire processes involved in the manufacture of hairpin stators as the predominant type for automotive traction applications.

Available online

3rd edition
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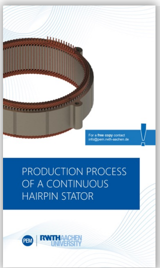
Ed.
PEM of RWTH Aachen University



Production Process of a Continuous Hairpin Stator

The “Production Process of a Continuous Hairpin Stator” guide presents continuous flat wire winding as an alternative to hairpin technology.

Available online

2nd edition
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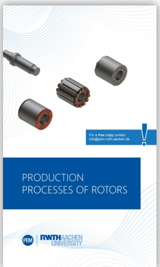

Production Processes of Rotors

The “Production Processes of Rotors” guide looks at the process chain for manufacturing rotors for automotive traction applications.

Available online

2nd edition
ISBN: 978-3-947920-63-1

Ed.
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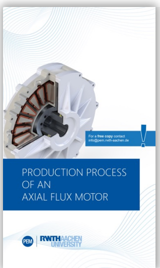


Production Process of an Axial Flux Motor

In the “Production Process of an Axial Flux Motor” guide, the process chain for manufacturing an axial flux motor is presented using a YASA design.

Available online

1st edition
ISBN: 978-3-947920-57-0

Ed.
PEM of RWTH Aachen University

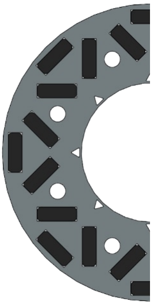
Radial Flux Machines

for automotive applications

General principle of operation

- In common radial flux machines, the rotor is usually arranged inside the stator as a so-called internal rotor.
- The stator windings are supplied with three-phase current and generate a rotating magnetic field.
- A magnetic field is also built up in the rotor, the properties of which differ depending on the excitation principle (PSM, SSM, IM – see below for explanations).
- Due to the interactions of the magnetic fields, the rotor follows the rotation of the stator magnetic field and rotates.
- In synchronous machines, the rotor speed is proportional to the rotational frequency of the stator field. Rotors of induction machines follow the exciter field with a delay.

Rotor Topologies



Permanently excited Synchronous Motor (PSM)

Exciter principle

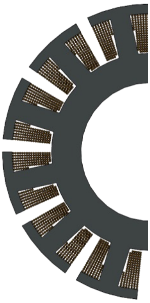
- Permanent magnets are mounted into or onto the rotor and generate a constant magnetic field.

Advantages

- Higher power density
- Higher efficiency

Disadvantages

- Higher material costs
- Increased assembly effort in magnetized state



Separately excited Synchronous Motor (SSM)

Exciter principle

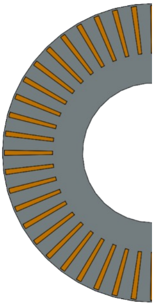
- The teeth of the rotor laminations are wound in the same way as the stator.
- The rotor winding is supplied with DC voltage via sliding contacts or inductively, and a constant electromagnetic field is generated.

Advantages

- Controllable rotor magnetic field
- Established production processes

Disadvantages

- Stochastic winding processes
- Mechanical wear of the sliding contacts



Induction Machine (IM)

Exciter principle

- Copper or aluminum bars are short-circuited to form a rotor cage.
- The rotating magnetic field of the stator induces current in the short-circuit cage, which ensures the build-up of an electromagnetic field.

Advantages

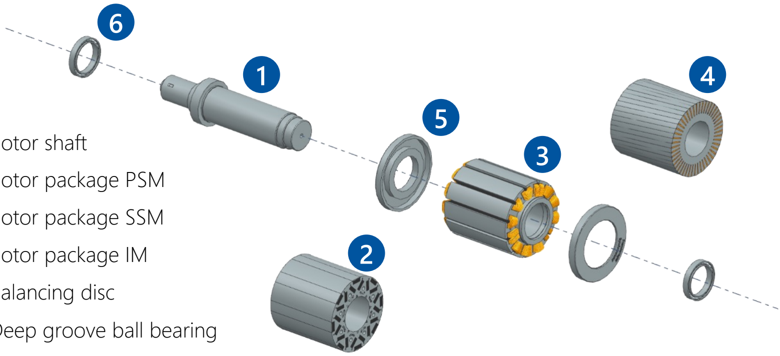
- No drag torque
- Compact process chain

Disadvantages

- Lower efficiency
- High process temperature in die casting

Design and Components

of rotors for radial flux machines



1. Rotor shaft
2. Rotor package PSM
3. Rotor package SSM
4. Rotor package IM
5. Balancing disc
6. Deep groove ball bearing

Process Chain

Rotor shaft	Rotor			EoL
	SSM	PSM	IM	
Shaping	Insulation	Magnet assembly	Rotor cage production	Rotor and shaft joining
Hardening	Winding	Magnet fixation		Balancing
Cleaning	Commutator assembly	Rotor sub-stack assembly (opt.)		
	Contacting	Magnetization		
	Impregnation			
	Testing			

Legend

- Process steps
- Sub-process steps

Further Information

Structure (additions)

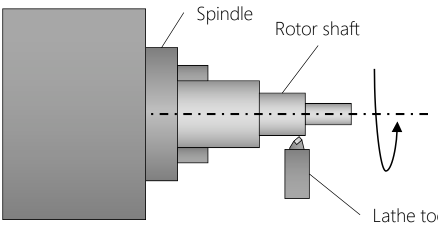
- Lamination stack made of electrical sheets
- Resolver mounted on shaft (position encoder/speed encoder)
- PSM: Magnets (plus adhesive or taping)
- SSM: Windings (coil), sliding contacts
- IM: Aluminum or copper cage
- Optional: Balancing disc, circlip
- Materials: Electrical sheet, enameled wire, copper, magnet material, insulation material

Quality requirements

- Magnetic field saturation and symmetry
- Torque ripple
- Freedom from damage (magnets, laminated core, bearings)
- "Noise – Vibration – Harshness" (NVH): Residual unbalance and balance quality
- External geometric and shape tolerances
- Cleanliness in production (freedom from chips)
- Small air gap (physically, at least approximately 0.2 millimeters are possible)

Rotor Shaft

Shaping and editing



Process Description

- In the manufacture of the rotor shaft, a shaft element (solid or hollow shaft) is given its shape by conventional metal-cutting turning. For functional surfaces, such as bearing seats, machining by grinding can also be performed.
- Functional elements such as the gearing for transmitting the rotor torque to the rest of the drive train are installed.
- The bearing seats are hardened, often by using the induction hardening process.
- If necessary, the surfaces are subsequently machined.
- Eventually, the rotor shaft is cleaned.

Further Information

Alternative technologies [excerpt]

- Forming by flowforming: Inductive heating and forming in the heated state, suitable for manufacturing the rotor as a hollow shaft
- Hollow or solid shaft design: Hollow shafts offer the potential of significant weight reduction and the possibility of cooling but are usually more cost-intensive to manufacture.

Quality features [excerpt]

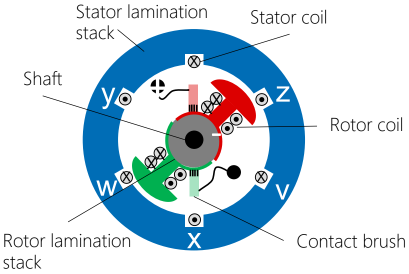
- High surface quality for tangential loads due to torque, and radial loads due to rotor weight
- Dimensional accuracy for error-free assembly
- Resistance to water-glycol mixtures and oil when using a rotor-internal cooling system

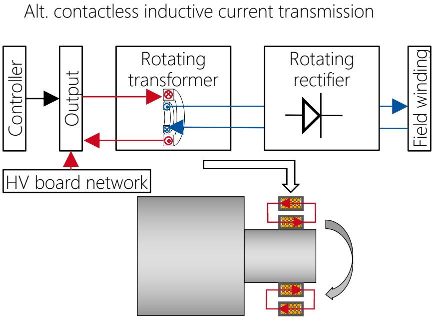
Quality impacts [excerpt]

- Shaft must be designed to be temperature-stable, as the rotor heats up during operation (max. 180°C)
- External force influences (tangential as well as radial) require high adherence to form tolerances (cylindricity, roundness, roundness and straightness)

Separately Excited Synchronous Machine

Design and components





Process Description

Insulate	Winding	Commutator	Contact	Check
Insulation of the manufactured poles	Winding of the poles (coil production)	Assembly of the commutator	Connection of the slip rings (commutator) that are connected to the DC source	Testing of the windings by test current or inductive test methods

Further Information

Design

- Generation of a rotating magnetic field by energizing the stator windings
- Generation of a constant magnetic field in the rotor by transmission of direct current

Operating principle

- Generation of a magnetic field by direct current (via slip ring)
- Energy conversion: Efficiency of 0.85 to 0.95

Advantages

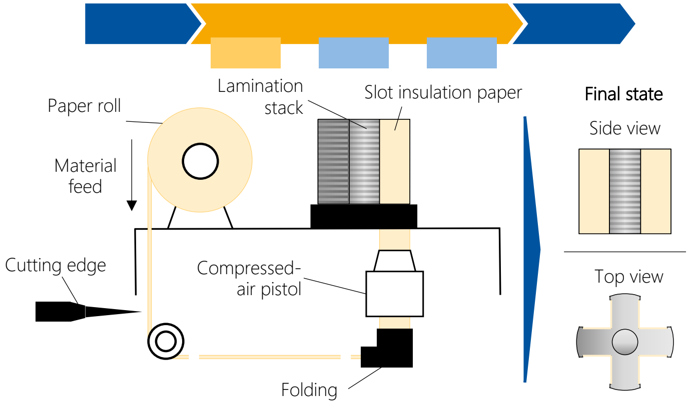
- No permanent magnets required, thus more cost-effective
- Simple construction
- Higher functional safety
- Good efficiency over operating range

Disadvantages

- A slip ring is needed to transmit the electrical energy, causing wear and abrasion in the air gap.
- Lower volumetric and gravimetric power density

SSM: Insulation

of the manufactured poles



Process Description

- The lamination stack must be electrically as well as mechanically separated from the winding to avoid electrical contact due to both material defects and assembly damage.
- The insulation paper or laminate is cut to the length of the lamination stack – with overhang, if necessary.
- The surface insulation material is folded according to the slot geometry and inserted into the rotor grooves.
- If necessary, the correct positioning of the paper is ensured by a slider.
- Additional insulation elements (cover caps, cover slides) are mounted after the winding has been inserted.

Further Information

Alternative technologies [excerpt]

- Overmolding
- Injection molding and assembly
- Powder coating

Quality features [excerpt]

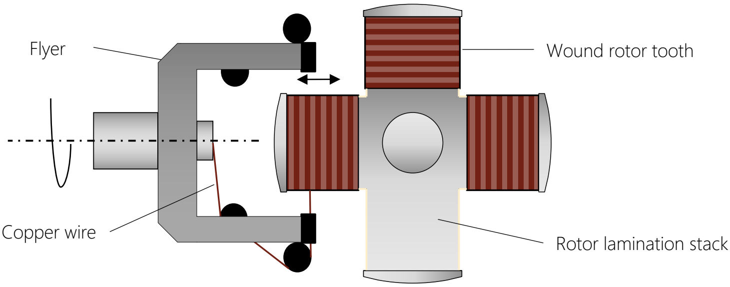
- Electrical insulation strength
- Mechanical filling level or wetting
- Overhang or overmolding of insulation material
- No damage during or after insertion
- Thermal conductivity and flexibility

Quality impacts [excerpt]

- Insertion speed/frequency/force
- Injection pressure, temperature, number of molds injected simultaneously
- Insulation material thickness

SSM: Winding

of the poles of the rotor lamination stack



Process Description

- In the production of separately excited synchronous machines, the coils for generating the magnetic field are wound onto the rotor poles.
- The enameled copper wire is fed through the so-called flyer with the help of wire clamps and placed on the rotor lamination stack's tooth to be wound.
- For even distribution, the roving frame is moved back and forth, and the wire is guided over guide jaws, if necessary.
- The wires are then cut, and the wire ends are fixed.
- Depending on the coil former's geometry, both concentrated and distributed windings are possible. Concentrated windings are common in rotors for traction applications.

Further Information

Alternative technologies [excerpt]

- Linear winding
- Needle winding
- Hairpin

Quality features [excerpt]

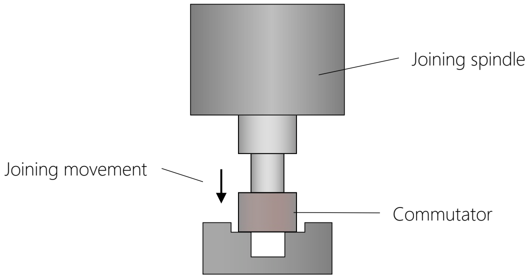
- Filling level (electrical/mechanical)
- Damage to the paint insulation of the wire
- Interconnection effort after completed winding

Quality impacts [excerpt]

- Wire tensile and wire bending forces
- Winding speed
- Traverse paths of the winding tool
- Rotation speed of the tool

SSM: Commutator

Assembly



Process Description

- The commutator, which enables continuous energization of the rotor, is joined to a shoulder of the rotor shaft in an axial joining process.
- The commutator is fixed to the rotor shaft by means of an interference fit created by pressing the oversize rotor shaft into the commutator.
- The power supply unit including the carbon brushes is usually screwed into the housing after the rotor and stator have been mounted.

Further Information

Alternative technologies [excerpt]

- Cold expansion by cooling the rotor shaft
- Bonding of the commutator to the rotor shaft
- Positive connection with feather key or serration

Quality features [excerpt]

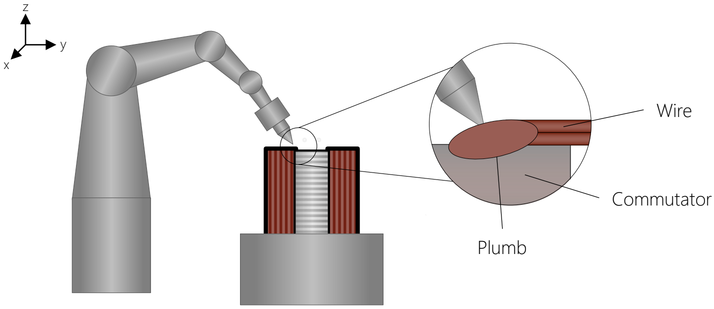
- Stable mechanical connection between commutator and rotor shaft
- High positioning accuracy and concentricity

Quality impacts [excerpt]

- Mechanical or thermal load on the rotor shaft
- Joining force
- Joining speed

SSM: Contacting

of the winding ends



Process Description

- Depending on the winding scheme, the first step is to make electrical contact between the individual winding strands, which may also be parallel.
- The winding and the commutator are electrically connected by soldering.
- Usually, end plates are mounted axially for mechanical fixation and protection of the winding.
- Contacting of the power supply unit with the power electronics, through which the commutator is supplied with current, takes place at the end of the motor assembly.

Further Information

Alternative technologies [excerpt]

- Clamping
- Laser welding
- Resistance welding
- (Hot) Crimping

Quality features [excerpt]

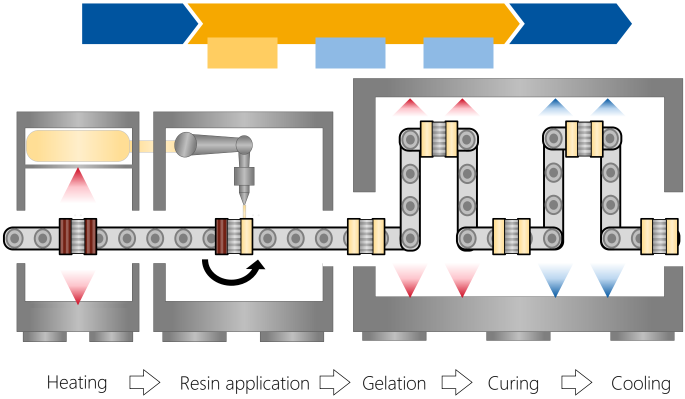
- Electrical resistance of the connection
- Low solder residue or weld spatter
- Mechanical strength of the connection
- Corrosion resistance

Quality impacts [excerpt]

- Temperature of the contact point
- Soldering temperature and time
- Energy input

SSM: Impregnation

of the windings



Process Description

- Rotor and resin are heated before application.
- The impregnating resin is applied to the windings.
- Slot penetration takes place according to the principle of the capillary effect.
- To ensure uniform distribution of the resin within the rotor, the rotor – including the pick-up – rolls continuously.
- Gelation, curing, and cooling take place in different plant sections and at individual temperature profiles.

Further Information

Alternative technologies [excerpt]

- Resin application: Trickling, roll dipping/rolling, hot dipping, vertical dipping, full potting
- Heat input: Convection, induction, current heat, infrared, ultraviolet (UV) radiation
- Additionally: Weight control before and after the process to determine the resin uptake; camera monitoring for application control

Quality features [excerpt]

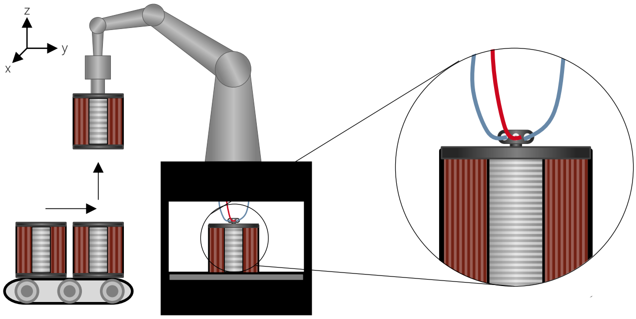
- High fill factor (especially in the slot)
- Avoidance of unfilled cavities
- High thermal conductivity
- Little rework required for cleaning
- Aging resistance
- Freedom from lamination stack damage

Quality impacts [excerpt]

- Process temperature profile
- Application quantity of the resin
- Resin temperature profile
- Rotation speed
- Continuity/synchronization of rotation
- Slot geometry

SSM: Electrical Test

of the rotor lamination stack with winding and interconnection



Process Description

- To ensure function, an electrical test of the lamination stack and winding is carried out.
- In the lamination stack, faults in the insulation generally lead to low motor efficiency due to increased (eddy current) losses or due to short circuits.
- Insulation testing of the lamination stack is carried out by using an external magnetic field, whereas measurement of the magnetic field strength after passing through the lamination stack is performed by using Hall sensors.
- Insulation testing of the winding is carried out, e.g., by means of a surge voltage test at different system boundaries (winding insulation, phase insulation, ground insulation) to detect insulation faults between conductors, phases, winding, and lamination stack.
- A resistance test by means of DC voltage measurement against ground is used to check the winding and contacting resistances.

Further Information

Alternative technologies [excerpt]

- Insulation test of the lamination stack
- Insulation test of the winding, surge voltage test
- Partial discharge test
- High-voltage test
- Resistance test

Quality features [excerpt]

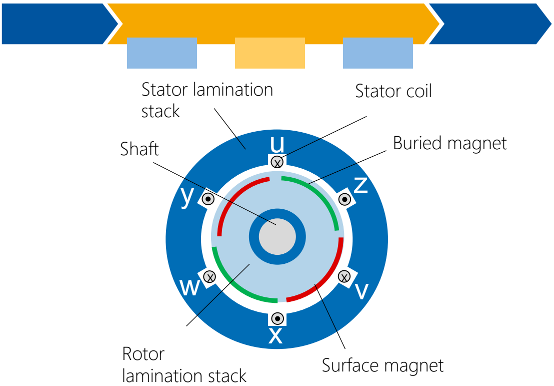
- Insulation strength lamination stack/ winding
- Insulation fault lamination stack/winding
- Winding and contact resistance

Quality impacts [excerpt]

- Material selection
- All upstream processes including handling

Permanently Excited Synchr. Machine

Design and components



Production Process

Magnet assembly	Magnet fixation	Magnetization
Magnetized or non-magnetized magnets are inserted into the slots provided for this purpose in the rotor lamination stack	Fastening of the permanent magnets mostly by means of a substance-to-substance connection	Application of a strong external magnetic field through a coil, resulting in irreversible alignment of magnetic spins in permanent magnents

Further Information

Design

- Generation of a rotating magnetic field by energizing the stator windings
- Generation of a temporally constant magnetic field by permanent magnets
- Commonly used magnetic raw material is Neodymium-Iron-Boron (NdFeB)

Operating principle

- Permanent magnetic field (due to permanent magnets)
- Energy conversion: Efficiency of 0.95 to 0.98

Advantages

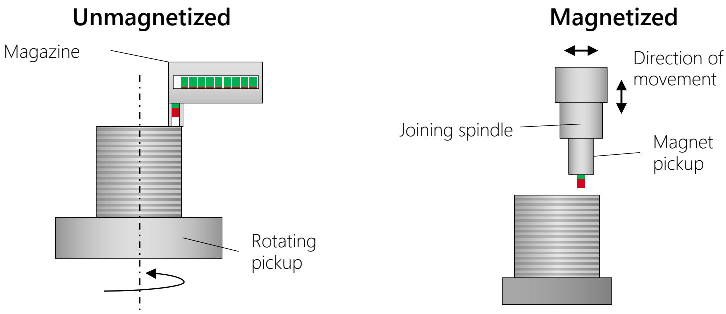
- Compact design with low weight thanks to high power density
- Greater range of electric vehicles due to higher efficiency with the same battery capacity

Disadvantages

- Required magnetic materials are limited resources
- Assembly of the magnets in magnetized state is time-consuming

PSM: Magnet Assembly

Insertion of the magnets



Process Description

- In preparation for the assembly of the magnets into the rotor lamination stack, the magnets are provided in an unmagnetized state in a magazine.
- The rotor lamination stack is fixed on a rotary table, and the first magnet slot is positioned under the feeder element.
- If necessary, an additive to fix the magnets in the lamination stack is applied to the magnets or in the slots of the lamination stack.
- The magnets are then deposited from the magazine into the slots, and the lamination stack is rotated to the next slot position.
- Eventually, the loaded rotor is removed automatically or manually.

Further Information

Alternative technologies [excerpt]

- Robotic direct assembly
- Joining with joining spindle

Quality features [excerpt]

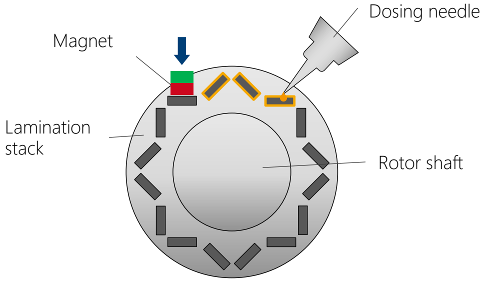
- High cycle rate
- High positioning accuracy
- Freedom from damage

Quality impacts [excerpt]

- Lower saturation during magnetization following assembly
- Increased effort when mounting already magnetized magnets

PSM: Magnet Fixation

Gluing-in of the magnets



Process Description

- An adhesive (1K or 2K) is applied manually or automatically to the magnet or applied to the slots of the rotor lamination stack.
- The magnets are inserted manually or by an automated joining unit into the slots of the rotor lamination stack.
- Curing takes place at temperatures between 20°C and 200°C, depending on the crosslinking mechanisms of the adhesive.
- By heating the adhesive, the process time can usually be significantly reduced with additional energy input.

Further Information

Alternative technologies [excerpt]

- Resin Transfer Molding
- Caulking
- Bandaging
- Expanding matrix
- Interference fit
- Tensioning and clamping elements

Quality features [excerpt]

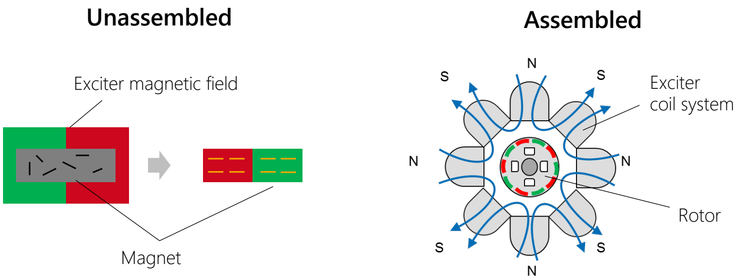
- Homogeneous and areal distribution of the adhesive
- High mechanical strength
- High running smoothness
- Temperature resistance
- Low energy input

Quality impacts [excerpt]

- Respective dosing system
- Adhesive residues in the machine system
- Temperature profile during curing

PSM: Magnetization

of the permanent magnets



Process Description

- Magnetization of the magnets is often carried out following assembly by using multi-coil exciter systems.
- The rotor (usually already with shaft) is placed in the exciter coil system, and an external magnetic field is applied so that the saturation flux density B_S is achieved in each magnet.
- For the best possible saturation of the magnets, with simple and near-surface magnet arrangements up to the saturation flux density, a variant-specific coil design is necessary.
- As soon as the necessary saturation is reached, the excitation field is reduced. A magnetic field with the material-specific "remanence B_R" flux density remains in the magnets.
- Eventually, the magnetized rotor is removed.

Further Information

Alternative technologies [excerpt]

- Placement of individual unmagnetized magnets in a coil assembly with a simple excitation magnetic field
- Application of the external magnetic field until the saturation flux density in the magnet material is reached
- Removal of the magnetized magnets and subsequent assembly in the rotor

Quality features [excerpt]

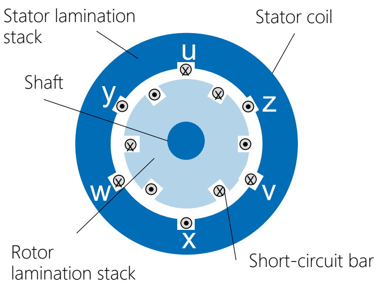
- Achieved remanence B_R
- Energy required to apply the necessary field strength for each magnet

Quality impacts [excerpt]

- Position of the magnets in the lamination stack
- Hysteresis behavior of the magnet material used

Induction Machine

Design and components



Production Process

Process alternative: Die casting	Process alternative: Welding
• Molten copper or aluminum is injected under pressure directly into the sheet package.	• Bar sections are mounted into the laminated core and connected with short-circuit rings to form a cage.

Further Information

Design

- Current is induced by the rotating field generated by the stator.
- Induced current creates a magnetic field that interacts with the stator's magnetic field to create torque.

Operating principle

- Generation of a temporary magnetic field by self-induction
- Energy conversion: Efficiency of 0.92 to 0.96

Advantages

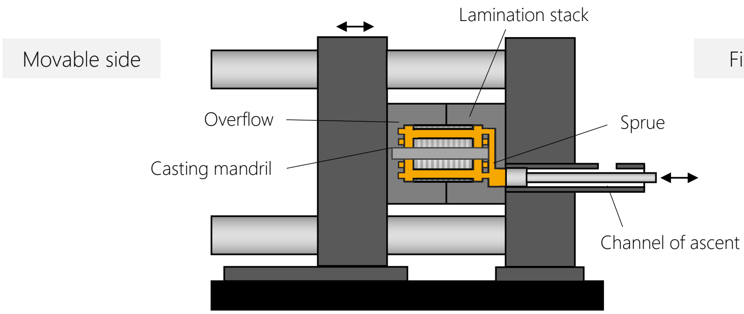
- Low cost and higher torque compared to DC motors
- No permanent magnets required
- Technically least complex while robust in operation

Disadvantages

- Lower power density and efficiency compared to PSM
- Increased heat development

IM: Die Casting

of the rotor cage



Process Description

- The lamination stack is placed on a casting mandril and pre-heated in a furnace.
- A solvent is applied to the warm mold (pre-heated or process heat).
- The lamination stack is inserted into the mold by robot, and the mold is closed.
- The channel of ascent is filled with the casting material (usually pure copper or aluminum).
- The clamping force is applied, the casting material is injected into the lamination stack via the channel of ascent, and the mold is held closed for a few seconds.
- The lamination stack with rotor cage is removed and cooled in an immersion tank, if necessary.
- The sprue is cut off by machining, the casting mandril is removed in a press, and the rotor shaft is inserted, if necessary.

Further Information

Alternative technologies [excerpt]

- Production of conductor rods and welding with short-circuit rings after assembly
- Production of semi-finished rotor cages (die casting) and welding with a short-circuit ring after assembly
- Soldering of the rodded rotor cage

Quality features [excerpt]

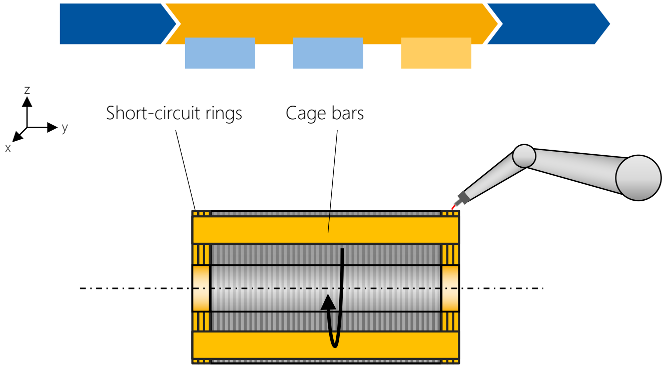
- Low porosity in the cast cage
- High mechanical strength of the rotor
- High temperature and wear resistance of the tool

Quality impacts [excerpt]

- Slot cross-section and length of the lamination stack as limiting factors for casting quality
- Stable process control during the casting process (shot speed, temperature of the melt, etc.)
- Casting material (copper or aluminum)

IM: Welding

of the rotor cage



Process Description

- From a bar profile (usually pure copper or aluminum), conductor rods are manufactured in the length of the lamination stack with a slight overhang.
- The short-circuit ring is built up in segments from individual discs produced in a stamping process.
- The short-circuit ring is applied to the end faces of the package, which can be done either directly during the packaging of the lamination stack or following the bar assembly.
- The lamination stack is clamped, and the conductor bars are inserted into slots provided for this purpose in the rotor lamination stack.
- The short-circuit rings are joined to the conductor bars by welding.

Further Information

Alternative technologies [excerpt]

- Die casting of the entire rotor cage (including the laminar squeeze casting process)
- Production of semi-finished rotor cages (die casting) and welding of a short-circuit ring after assembly
- Soldering the rodded rotor cage

Quality features [excerpt]

- Low porosity in the conductors
- Good electrical contacting of the short-circuit ring
- High electrical slot fill factor
- Requirement-oriented composition of the short-circuit ring (mechanical strength, conductivity, etc.)

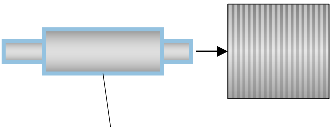
Quality impacts [excerpt]

- Cleanliness of contact points before the welding process
- Welding process control
- Thermal influence due to the joining process

Joining

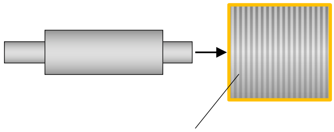
of the shaft into the rotor pack

Stretching



Cooled rotor shaft

Heat shrinking



Heated lamination stack

Process Description

- The rotor is mounted on the rotor shaft so that it cannot rotate, allowing the generated torque to be transmitted to the shaft.
- In most cases, the frictional connection between the lamination stack and the rotor shaft is produced by thermal joining (stretching or heat shrinking).
  - Stretching: The rotor shaft is cooled in a nitrogen bath (temperature: below -196°C) so that the shaft's volume is reduced, and the shrunken shaft can be joined into the lamination stack.
  - Heat shrinking: The lamination stack is usually heated inductively to a temperature of around 200°C so that the lamination stack expands thermally and can be joined onto the rotor shaft.
- The subsequent temperature compensation between rotor shaft and lamination stack results in a shaft-hub connection by press fit.

Further Information

Alternative technologies [excerpt]

- Joining by frictional connection: Pressing the shaft into the lamination stack creates a frictional connection between the rotor shaft and the lamination stack.
- Joining with the aid of a feather key: Positive connection by mounting a feather key
- Joining by bonding: The rotor laminations are glued to the shaft. The adhesive becomes warm during the chemical reaction.
- Joining by welding: The shaft is inserted into the rotor laminations. Both components are then circumferentially welded.

Quality features [excerpt]

- Little change in material properties
- High mechanical stability of the connection
- Compliance with shape and position tolerances

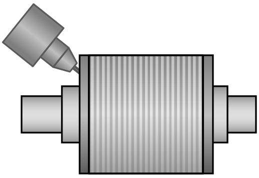
Quality impacts [excerpt]

- Heating or shrinkage can lead to structural changes and irregular warpage
- No damage-free disassembly
- Consideration of axial securing during join-ing with the aid of a feather key

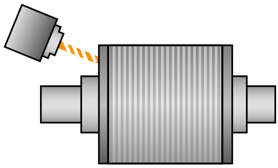
Balancing

of the entire rotor

Additive



Subtractive



Process Description

- To prevent bearing damage and minimize noise and vibration, the entire rotor system is balanced in the final process step.
- The existing unbalance, both static and dynamic, is detected via an unbalance measurement.
- The unbalance is corrected by applying or removing mass to or from the rotor system at balancing discs provided for this purpose or directly at the lamination stack.
- Depending on the application and operating speed, different balancing grades have to be achieved (DIN ISO1940-1:2004-04).

Further Information

Alternative technologies [excerpt]

- Additive balancing: Spot mass is **added** to the overall system at defined points (example: build-up welding).
- Subtractive balancing: Mass is **removed** from the entire system at defined points (example: holes drilled in the balancing disc or in the lamination stacks).

Quality features [excerpt]

- Unbalance (mass and radius)
- Position of the unbalance
- Rotational speed
- Eccentricity
- Low weight for compensation

Quality impacts [excerpt]

- Targeted balancing qualities
- Dynamic or static unbalances
- Mass of the unbalance

Radial Flux Machines

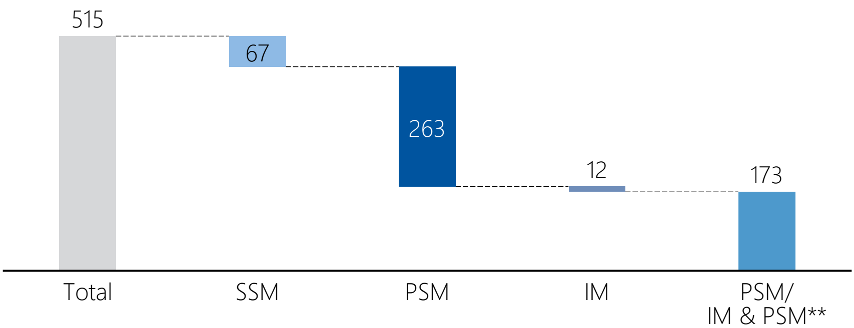
in automotive applications

Current market overview

- An analysis of the vehicle market in Germany shows that, at present, permanently excited synchronous machines are primarily used.
- The system output of the models currently available in Germany or announced for 2024 varies between 33 and 828 kilowatts.
- The use of multiple electric motors per vehicle is becoming increasingly established.
- By using different engine designs in one vehicle, topology-specific properties can be combined with each other.

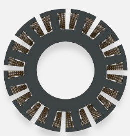
Motor Topologies in Use

Registration numbers for vehicle models available in Germany distributed across the engine topologies (in thousand units)*



Use of SSMs

- Current market share: 13%
- SSMs can be used to avoid rare-earth metals at slightly reduced efficiency compared to PSMs.

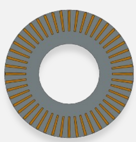


Use of PSMs

- Current market share: 51%
- The high power density and high efficiency make compact and powerful motor designs possible.

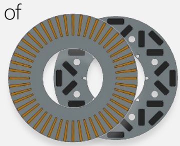
Use of IMs

- Current market share: 2%
- IMs are free of rare-earth metals and are a low-cost alternative for high volumes with high efficiency in the high-speed range.



Use of PSMs/IMs & PSMs**

- Current market share: 34%
- By combining IMs and PSMs, the properties of the two topologies can be used in a complementary manner.



* Distribution of electric motor topologies of passenger cars registered in Germany in 2023 (own research from July 2024)
** Allocation of the model topologies according to performance

PEM of RWTH Aachen University

The Chair of Production Engineering of E-Mobility Components (PEM) of RWTH Aachen University was founded in 2014 by StreetScooter co-inventor Professor Achim Kampker. In numerous research groups, the team is dedicated to all aspects of the development, production, and recycling of battery systems and their components as well as hydrogen technologies, the production of the electric powertrain and entire vehicle concepts. PEM's focus is always on sustainability and cost reduction – with the goal of a seamless Innovation Chain from basic research to large-scale production in the immediate vicinity.

PEM's "Electric Drive Production" research group contributes to the economic, variant-flexible, future-oriented, and sustainable production of the electric drive and its active components (rotor and stator). The focus is on the investigation of issues along the entire value chain – from the semi-finished product to the finished drive and from the individual process to the holistic consideration of cross-process interactions. In line with the PEM philosophy, the team works on innovative technologies, always keeping an eye on the path to industrial application and solution scaling.

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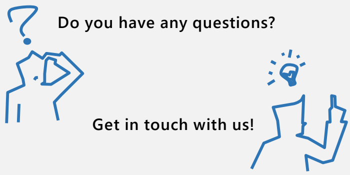

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